

Effects of cigarette smoke on the human airway epithelial cell transcriptome

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Cigarette smoke is the major cause of lung cancer, the leading cause of cancer death, and of chronic obstructive pulmonary disease, the fourth leading cause of death in the United States. Using high-density gene expression arrays, we describe genes that are normally expressed in a subset of human airway epithelial cells obtained at bronchoscopy (the airway transcriptome), define how cigarette smoking alters the transcriptome, and detail the effects of variables, such as cumulative exposure, age, sex, and race, on cigarette smoke-induced changes in gene expression. We also determine which changes in gene expression are and are not reversible when smoking is discontinued. The persistent altered expression of a subset of genes in former smokers may explain the risk these individuals have for developing lung cancer long after they have discontinued smoking. The use of gene expression profiling to explore the normal biology of a specific subset of cells within a complex organ across a broad spectrum of healthy individuals and to define the reversible and irreversible genetic effects of cigarette smoke on human airway epithelial cells has not been previously reported.

Approximately 1.25 billion people smoke cigarettes daily worldwide (1). Cigarette smoking is responsible for 90% of all lung cancers, the leading cause of cancer deaths in the United States and the world (2, 3). Smoking is also the major cause of chronic obstructive pulmonary disease (COPD), the fourth leading cause of death in the United States (4). Despite the well established causal role of cigarette smoking in lung cancer and COPD, only 10–20% of smokers actually develop these diseases (5). Few indicators of which smokers are at highest risk for developing either lung cancer or COPD exist, and it is unclear why individuals remain at high risk decades after they have stopped smoking (6).

Given the burden of lung disease created by cigarette smoking, surprisingly few studies (7, 8) have been done in humans to determine how smoking affects the epithelial cells of the pulmonary airways that are exposed to the highest concentrations of cigarette smoke or what smoking-induced changes in these cells are reversible when subjects stop smoking. With the two exceptions noted above, which examine a specific subset of genes in humans, studies investigating the effects of tobacco on airway epithelial cells have been in cultured cells, in human alveolar lavage samples in which alveolar macrophages predominate, or in rodent smoking models [summarized by Gebel *et al.* (9)]. Several recent studies have used DNA microarray technology to study normal and cancerous whole lung tissue and have identified molecular profiles that distinguish the various subtypes of lung cancer and predict clinical outcome in a subset of these patients (10–13).

Based on the concept that genetic alterations in airway epithelial cells of smokers represent a “field defect” (14, 15), we obtained human epithelial cells at bronchoscopy from brushings of the right main bronchus proximal to the right upper lobe of the lung and defined profiles of gene expression in these cells using the U133A GeneChip array (Affymetrix, Santa Clara, CA). We describe the subset of genes expressed in large-airway

epithelial cells (the airway transcriptome) of healthy never smokers, thereby gaining insights into the biological functions of these cells. We found a large number of genes whose expression is altered by cigarette smoking, defined genes whose expression correlates with cumulative pack-years of smoking, and identified genes whose expression does and does not return to normal when subjects discontinue smoking. In addition, we found a subset of smokers who were “outliers,” expressing some genes in a fashion that differed from most smokers. One of these “outliers” developed lung cancer within 12 months of expression profiling, suggesting that gene expression profiles of smokers with cancer may differ from those of smokers without lung cancer.

Materials and Methods

Study Population and Sample Collection. We recruited nonsmoking and smoking subjects ($n = 93$) to undergo fiberoptic bronchoscopy at Boston Medical Center between November 2001 and June 2003. Nonsmoking volunteers with significant environmental cigarette exposure and subjects with respiratory symptoms or regular use of inhaled medications were excluded. For each subject, a detailed smoking history was obtained including number of pack-years, number of packs per day, age started, age quit, and environmental tobacco exposure.

Bronchial airway epithelial cells were obtained from brushings of the right mainstem bronchus taken during fiberoptic bronchoscopy with an endoscopic cytobrush (Cellebity Endoscopic Cytology Brush, Boston Scientific, Boston). The brushes were immediately placed in TRIzol reagent (Invitrogen) after removal from the bronchoscope and kept at -80°C until RNA isolation was performed. RNA was extracted from the brushes by using TRIzol Reagent (Invitrogen) according to the manufacturer's protocol, with a yield of 8–15 μg of RNA per patient. Integrity of the RNA was confirmed by running it on a RNA-denaturing gel. Epithelial cell content of representative bronchial brushing samples was quantified by cytocentrifugation (Cytospin, ThermoShandon, Pittsburgh) of the cell pellet and staining with a cytokeratin antibody (Signet Laboratories, Dedham, MA). The study was approved by the Institutional Review Board of Boston University Medical Center, and all participants provided written informed consent.

Microarray Data Acquisition and Preprocessing. We obtained a sufficient quantity of good-quality RNA for microarray studies from 85 of the 93 subjects recruited into our study. Six to eight micrograms of total RNA was processed, labeled, and hybridized to Affymetrix HG-U133A GeneChips containing $\approx 22,500$ human transcripts (for detailed protocol, see *Supporting Text*, which

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Data deposition: The data reported in this paper have been deposited in National Center for Biotechnology Information's Gene Expression Omnibus (accession no. GSE994) and at <http://pulm.bumc.bu.edu/aged>.

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is published as supporting information on the PNAS web site). A single weighted mean expression level for each gene was derived by using MICROARRAY SUITE 5.0 software (Affymetrix). Using a one-sided Wilcoxon signed-rank test, the MAS 5.0 software also generated a detection P value [$P_{(\text{detection})}$ value] for each gene that indicates whether the transcript was reliably detected. We scaled the data from each array to normalize the results for interarray comparisons. Microarray data normalization was accomplished in MAS 5.0, where the mean intensity for each array (top and bottom 2% of genes excluded) was corrected (by a scaling factor) to a set target intensity of 100.

Arrays of poor quality were excluded based on several quality-control measures. Each array's scanned image was required to be free of any significant artifacts, and the bacterial genes spiked into the hybridization mix had to have a $P_{(\text{detection})}$ value <0.05 (called present). If an array met these criteria, it was evaluated based on three other quality measures: the 3' to 5' ratio of the intensity for GAPDH, the percent of genes detected as present, and the percent of "outlier" genes as determined by a computational algorithm we developed (see supporting information for details).

In addition to the set of rules above, one further quality control measure was applied to each array. Although cytokeratin stains of selected specimens reveal that $\approx 90\%$ of nucleated cells are epithelial, we developed a gene filter to exclude specimens potentially contaminated with inflammatory cells. A group of genes on the U133A array was identified that should be expressed in bronchial epithelial cells and a list of genes that are specific for various lineages of white blood cells and distal alveolar epithelial cells (see Tables 2 and 3, which are published as supporting information on the PNAS web site). Arrays whose 90th percentile for the $P_{(\text{detection})}$ value was >0.05 for genes that should be detected in epithelial cells or whose 80th percentile $P_{(\text{detection})}$ value was <0.05 for genes that should not be expressed in bronchial epithelial cells were excluded from the study. Ten of the 85 samples were excluded based on the quality-control filter and the epithelial content filter described above.

In addition to filtering out poor-quality arrays, we applied a gene filter to remove genes that were not reliably detected. From the complete set of $\approx 22,500$ probe sets on the U133 array, we filtered out probe sets whose $P_{(\text{detection})}$ value was not <0.05 in at least 20% of all samples. A total of 9,968 probe sets passed our filter and were used in all further statistical analyses for the data set.

Microarray Data Analysis. Clinical information, array data, and gene annotations are stored in an interactive MYSQL database coded in PERL available at <http://pulm.bumc.bu.edu/aged/index.html>. All statistical analyses below and within the database were performed with R V. 1.6.2 software (available at <http://r-project.org>). The gene annotations used for each probe set were from the October 2003 NetAffx HG-U133A Annotation Files.

Technical, spatial (right and left bronchus from same subject), and temporal (baseline and at 3 months from same subject) replicates were obtained from selected subjects for quality control. Pearson correlations were calculated for technical, spatial, and temporal replicate samples from the same individual (see supporting information for details). An unsupervised analysis of the microarray data was performed by hierarchical clustering the top 1,000 most variable probe sets (determined by coefficient of variation) across all samples with log-transformed z -score normalized data. The analysis was performed by using a Pearson correlation (uncentered) similarity metric and average linkage clustering with CLUSTER and TREEVIEW software obtained at <http://rana.lbl.gov/EisenSoftware.htm> (see Fig. 4, which is published as supporting information on the PNAS web site).

The normal large-airway transcriptome was defined by the genes whose median $P_{(\text{detection})}$ value was <0.05 across all 23 healthy never smokers (7,119 genes expressed across the majority of subjects) and a subset of these 7,119 genes whose $P_{(\text{detection})}$ value was <0.05 in all 23 subjects (2,382 genes expressed across all subjects). The coefficient of variation for each gene in the transcriptome was calculated as the standard deviation divided by the mean expression level multiplied by 100 for that gene across all nonsmoking individuals. To identify functional categories that were over- or underrepresented within the airway transcriptome, GOMINER software (16) was used to functionally classify the genes expressed across all nonsmokers (2,382 probe sets) by the molecular function categories within gene ontology. Multiple linear regressions were performed on the top 10% most variable probe sets (712 probe sets, as measured by the coefficient of variation) in the normal airway transcriptome (7,119 probe sets) to study the effects of age, gender, and race on gene expression (see supporting information).

To examine the effect of smoking on the airway, a two-sample Student t test was used to test for genes differentially expressed between current smokers ($n = 34$) and never smokers ($n = 23$). To quantify how well a given gene's expression level correlates with the number of pack-years of smoking among current smokers, Pearson correlation coefficients were calculated (see supporting information). For multiple comparison correction, a permutation test was used to assess the significance of our P value threshold for any given gene's comparison between two groups [$P_{(t \text{ test})}$ value] or between a clinical variable [$P_{(\text{correlation})}$ value] (see supporting information for details). To further characterize the behavior of current smokers, 2D hierarchical clustering of all never smokers and current smokers using the genes that were differentially expressed between current vs. never smokers was performed. Hierarchical clustering of the genes and samples was performed by using log-transformed z -score normalized data with a Pearson correlation (uncentered) similarity metric and average linkage clustering with CLUSTER and TREEVIEW software.

Multidimensional scaling and principal component analysis were used to characterize the behavior of former smokers ($n = 18$) based on the set genes differentially expressed between current and never smokers by using PARTEK 5.0 software (Partek, St. Charles, MO). In addition, we executed an unsupervised hierarchical clustering analysis of all 18 former smokers according to the expression of the genes differentially expressed between current and never smokers. To identify genes irreversibly altered by cigarette smoking, we performed Student's t test between former smokers ($n = 18$) and never smokers ($n = 23$) across the genes that were considered differentially expressed between current and never smokers (see supporting information for details).

Given the invasive nature of the bronchoscopy procedure, we were unable to recruit age-, race-, and gender-matched patients for the smoker vs. nonsmoker comparison. Because of baseline differences in age, gender, and race between never- and current-smoker groups (see Table 4, which is published as supporting information on the PNAS web site), we performed an analysis of covariance (ANCOVA) to test the effect of smoking status (never or current) on gene expression while controlling for the effects of age (the covariate). In addition, a two-way ANOVA was performed to test the effect of smoking status (never or current) on gene expression while controlling for the fixed effects of race (encoded as three racial groups: Caucasian, African American, and other) or gender and the interaction terms of status:race or status:gender. Both the ANCOVA and two-way ANOVA were performed with PARTEK 5.0 software.

Quantitative PCR Validation. Quantitative real-time PCR was used to confirm the differential expression of a select number of

genes. Primer sequences were designed with PRIMER EXPRESS software (Applied Biosystems). Forty cycles of amplification, data acquisition, and data analysis were carried out in an ABI Prism 7700 Sequence Detector (Applied Biosystems). All real-time PCR experiments were carried out in triplicate on each sample (see supporting information for protocol details).

Additional Information. Additional information from this study, including the raw image data from all microarray samples (.DAT files), expression levels for all genes in all samples (stored in a relational database), user-defined statistical and graphical analysis of data, and clinical data on all subjects are available at <http://pulm.bumc.bu.edu/aged>. Data from our microarray experiments have also been deposited in National Center for Biotechnology Information Gene Expression Omnibus (accession no. GSE994).

Results and Discussion

Study Population and Replicate Samples. Microarrays from 75 subjects passed the quality-control filters described above and are included in this study. Demographic data on these subjects, including 23 never smokers, 34 current smokers, and 18 former smokers, are presented in Table 4. Bronchial brushings yielded 90% epithelial cells, as determined by cytokeratin staining, with the majority being ciliated cells. Samples taken from the right and left main bronchi in the same individual were highly reproducible with an R^2 value of 0.92, as were samples from the same individual taken 3 months apart with an R^2 value of 0.85 (see Fig. 5 and Table 5, which are published as supporting information on the PNAS web site).

The Normal Airway Transcriptome. A total of 7,119 genes were expressed at measurable levels in the majority of never smokers, and 2,382 genes were expressed in all 23 healthy never smokers. Expression levels of the 7,119 genes varied relatively little; 90% had a coefficient of variation (standard deviation from the mean) of <50% (see Fig. 6, which is published as supporting information on the PNAS web site). Only a small part of the variation between subjects could be explained by age, gender, or race on multiple linear regression analysis (see Table 6, which is published as supporting information on the PNAS web site).

Table 1 depicts the GOMINER molecular functions (16) of the 2,382 genes expressed in large-airway epithelial cells of all healthy never smokers. Genes associated with oxidant stress, ion and electron transport, chaperone activity, vesicular transport, ribosomal structure, and binding functions are overrepresented. Genes associated with transcriptional regulation, signal transduction, pores and channels, and immune, cytokine, and chemokine genes are underrepresented. Upper airway epithelial cells, at least in healthy subjects, appear to serve as an oxidant and detoxifying defense system for the lung but serve few other complex functions in the basal state.

Effects of Cigarette Smoking on the Airway Transcriptome. Smoking altered the airway epithelial cell expression of numerous genes. Ninety-seven genes were found to be differentially expressed by Student's t test between current and never smokers at $P < 1.06 \times 10^{-5}$. This $P_{(t \text{ test})}$ value threshold was selected based on a permutation analysis performed to address the multiple comparison problem inherent in any microarray analysis (see supporting information for further details). We chose a very stringent multiple-comparison correction and $P_{(t \text{ test})}$ value threshold to identify a subset of genes altered by cigarette smoking with only a small probability of having a false positive. Of the 97 genes that passed the permutation analysis, 68 (73%) represented increased gene expression among current smokers. The greatest increases were in genes that coded for xenobiotic functions such as CYP1B1 (30-fold) and DBDD (5-fold), antioxidants such as

Table 1. GOMINER molecular functions of genes in airway epithelial cells

Molecular functions	Overrepresented (cells/array)	Underrepresented (cells/array)
Binding activity		
RNA binding	0.76 (273/366)	
Translation	0.72 (72/101)	
Transcription		0.30 (214/704)
GTP binding	0.55 (106/194)	
GTPase	0.55 (83/152)	
G nucleotide	0.52 (128/246)	
Receptor		0.20 (79/396)
Chaperone	0.62 (80/119)	
Chemokine		0.24 (10/42)
Cytokine		0.20 (39/194)
Enzyme activity	0.46 (1346/2925)	
Oxidoreductase	0.54 (225/417)	
Isomerase	0.56 (48/82)	
Signal transduction		0.29 (490/1716)
Structural	0.46 (253/548)	
Transcription regulator		0.35 (321/917)
Transporter		
Carrier	0.48 (175/363)	
Ion	0.56 (130/231)	
Anion		0.26 (15/61)
Cation	0.64 (116/180)	
Metal	0.68 (42/62)	
Electron	0.58 (131/226)	
Channel/pore		0.16 (43/269)

Major molecular functional categories and subcategories of 2,382 genes expressed in all never smoker subjects. Over- or underrepresentation of categories are determined by using Fisher's exact test. The null hypothesis is that the number of genes in our flagged set belonging to a category divided by the total number of genes in the category is equal to the number of flagged genes *not* in the category divided by the total number of genes *not* in the category. Equivalency in these two proportions is consistent with a random distribution of genes into functional categories and indicates no enrichment or depletion of genes in the category being tested. Categories considered to be statistically [$P_{(GO)} < 0.05$] over- or underrepresented by GOMINER are shown. Cells/arrays refers to the ratio of the number of genes expressed in epithelial cells divided by the number of genes on the U133A array in each functional category. Actual numbers are in parentheses.

GPX2 (3-fold) and ALDH3A1 (6-fold), and genes involved in electron transport such as NADPH (4-fold). In addition, several cell adhesion molecules, including CEACAM6 (2-fold) and claudin 10 (3-fold), were increased in smokers, perhaps in response to the increased permeability that has been found upon exposure to cigarette smoke (17). Genes that decreased included TU3A (4-fold), MMP10 (2-fold), HLF (2-fold), and CX3CL1 (2-fold). In general, genes that were increased in smokers tended to be involved in regulation of oxidant stress and glutathione metabolism, xenobiotic metabolism, and secretion. Expression of several putative oncogenes (pirin, CA12, and CEACAM6) were also increased. Genes that decreased in smokers tended to be involved in regulation of inflammation, although expression of several putative tumor suppressor genes (TU3A, SLIT1 and -2, and GAS6) were decreased. Changes in the expression of select genes were confirmed by real-time RT-PCR (see Fig. 7, which is published as supporting information on the PNAS web site).

Fig. 1 shows 2D hierarchical clustering of all the current and never smokers based on the 97 genes that are differentially expressed between the two groups (tree for genes not shown). The expression of a subset of genes in three current smokers (patients 56, 147, and 164) was similar to that of never smokers.

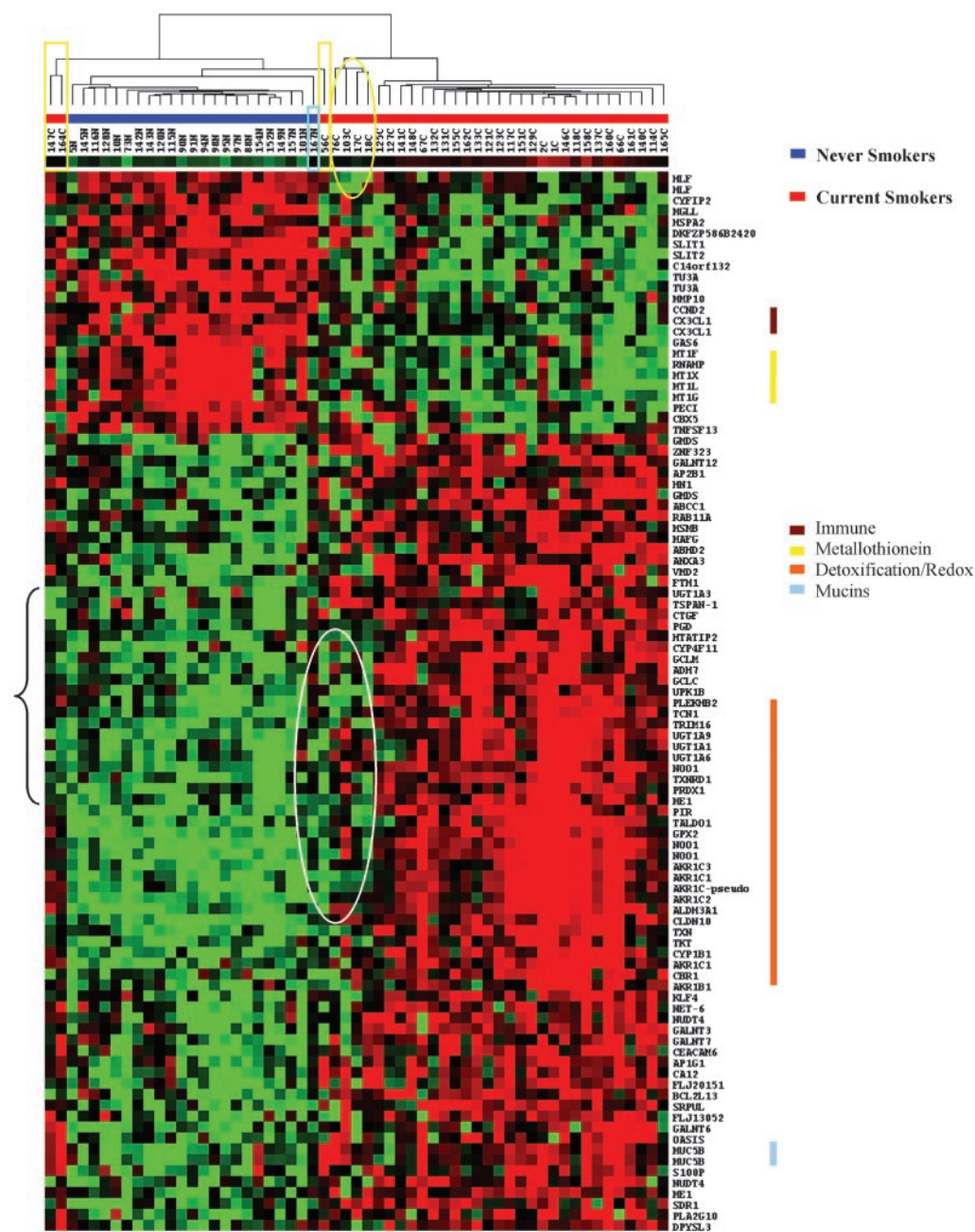


Fig. 1. Clustering of current- and never-smoker samples. Hierarchical clustering of current ($n = 34$) and never ($n = 23$) smokers according to the expression of the 97 genes differentially expressed between current and never smokers. Although current and never smokers separate into two groups, three current smokers appear to cluster with never smokers (yellow rectangle). Expression of several redox-related and xenobiotic genes in these subjects was not increased (brackets) and therefore resembled that of never smokers despite substantial smoke exposure. Also, a subset of current smokers (yellow circle) did not up-regulate expression of a number of predominantly redox/xenobiotic genes (white circle) to the same degree as other smokers. In addition, a never smoker, 167N (blue box), is an outlier among never smokers and expresses a subset of genes at the level of current smokers. HUGO gene ID listed for all 97 genes. The functional classification of select genes is shown. Red, high level of expression; green, low level of expression; black, mean level of expression.

These three smokers, who were similar clinically to other smokers, also segregated in the same fashion when clusters were based on the top 361 genes differentially expressed between never and current smokers ($P < 0.001$). Expression of a number of redox-related and xenobiotic genes was not increased in these three smokers (subjects 147C, 164C, and 56C), and, therefore, their profile resembled that of never smokers despite their substantial and continuing exposure to cigarette smoke. Thus, these individuals failed to increase expression of a number of genes that serve as protective detoxification and antioxidant genes, potentially putting them at risk of more severe smoking-related damage. Whether these differences represent genetic polymorphisms, and whether these individuals represent the 10–15% of smokers who ultimately develop lung cancer is uncertain. However, one of these subjects (subject 147C) subsequently developed lung cancer during the 1-year follow-up, suggesting some

link between the divergent patterns of gene expression and presence of or risk for developing lung cancer. Also, a subset of four additional current smokers clustered with current smokers but did not up-regulate expression of a cluster of predominantly redox/xenobiotic genes to the same degree as other smokers, although none of these smokers had developed lung cancer in 6 months of follow-up. In addition, a never smoker (subject 167N) is an outlier among never smokers and expresses a subset of genes at the level of current smokers. We reviewed this subject's clinical history and were unable to identify any obvious environmental exposures (i.e., second-hand smoke exposure) that might explain the divergent pattern of gene expression.

As might be expected, changes in gene expression were also correlated with cumulative cigarette exposure (pack-years). Although 159 and 661 genes correlated with cumulative smoking history at $P < 0.001$ and $P < 0.01$ levels, respectively (see Table

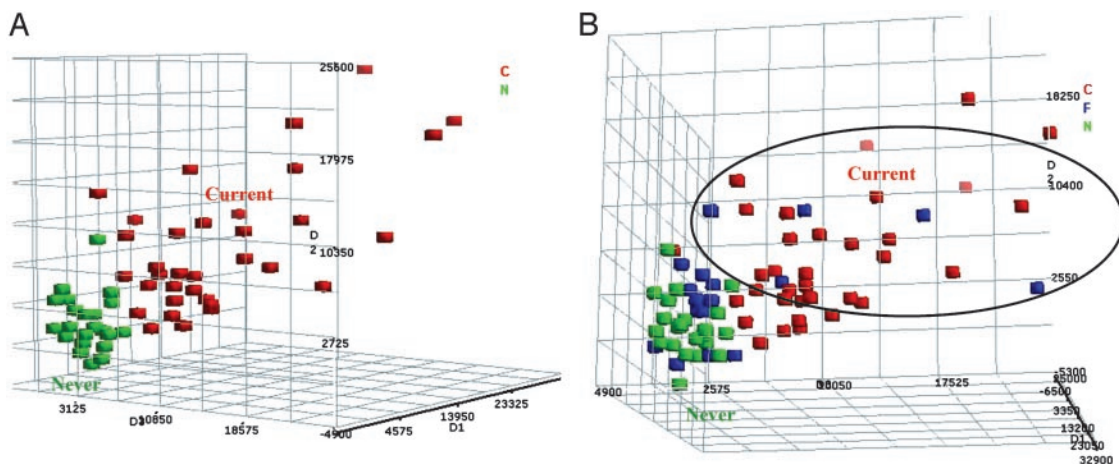


Fig. 2. Multidimensional scaling plot of current-, never-, and former-smoker samples. Multidimensional scaling plot of current (red boxes), never (green boxes), and former smokers (blue boxes) in 97 dimensional space according to the expression of the 97 genes differentially expressed between current and never smokers. Current and never smokers separate into their two classes according to the expression of these genes (A). When former smokers are plotted according to the expression of these genes (B), a majority of former smokers appear to group more closely to never smokers. However, a number of former smokers group more closely to current smokers (black circle). The only clinical variable that differed between the two groups of former smokers was length of smoking cessation ($P < 0.05$), with former smokers who quit within 2 years clustering with current smokers. The multidimensional scaling plots are reduced-dimension representations of the data, and the axes on the figure have no units.

7, which is published as supporting information on the PNAS web site), only five genes correlated with pack-years at the $P < 3.1 \times 10^{-6}$ threshold (based on permutation analysis; see supporting information for details). These genes include cystatin, which has been shown to correlate with tumor growth and inflammation (18); HBP17, which has been shown to enhance fibroblast growth factor activity (19); and BRD2, which is a transcription factor that acts with E2F proteins to induce a number of cell cycle-related genes (20). Among the genes that were correlated at the $P < 0.0001$ level, a number of genes decreased with increasing cumulative smoking history, including genes that are involved in DNA repair (RPA1).

Because of baseline differences in age, sex, and race between never- and current-smoker groups, analysis of covariance and two-way ANOVA were performed to test the effect of smoking status on gene expression while controlling for the effects of age, gender, race, and two-way interactions. Many of the genes found to be modulated by smoking in this analysis were also found by

using the simpler Student t test. Age and gender had little effect on gene expression changes induced by smoking, whereas race appeared to influence the effect of smoking on the expression of a number of genes. The ANOVA controlling for race yielded 16 genes not included in the set of 97 genes differentially expressed between current and never smokers (see Table 8, which is published as supporting information on the PNAS web site). Given the relatively small sample size for this subgroup analysis, these observations must be confirmed in a larger study but may account in part for the reported increased incidence of lung cancer in African-American cigarette smokers (21).

Thus, the general effect of smoking on large-airway epithelial cells was to induce expression of xenobiotic metabolism and redox stress-related genes and to decrease expression of some genes associated with regulation of inflammation. Several putative oncogenes were up-regulated and tumor suppressor genes were down-regulated, although their roles in smoking-induced lung cancer remain to be determined. Risk for developing lung

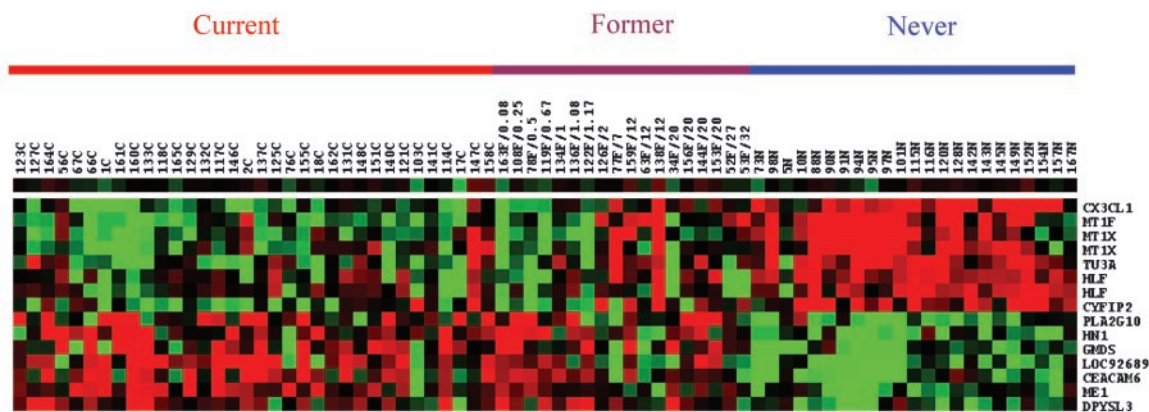


Fig. 3. Genes irreversibly altered by cigarette smoke. Hierarchical clustering plot of 15 of the 97 probe sets from Fig. 1 that remain differentially expressed between former vs. never smokers ($P < 0.0001$) as long as 30 years after cessation of smoking. Samples are grouped according to smoking status and length of smoking cessation (samples are not being clustered and, thus, no dendrogram occurs on the sample axis). Patient ID, status (C, F, or N), and length of time (years) because smoking cessation are shown for each sample. Current, current smokers; former, former smokers; never, never smokers. HUGO gene ID is listed for all 15 genes. Two genes (HLF and MT1X) appear twice in the analysis (i.e., two different probe sets corresponding to the same gene). Red, high level of expression; green, low level of expression; black, mean level of expression.

cancer in smokers has been shown to increase with cumulative pack-years of exposure (22), and a number of putative oncogenes correlate positively with pack-years, whereas putative tumor suppressor genes correlate negatively.

It is unlikely that the alterations we observed in smokers were caused by a change in cell types obtained at bronchoscopy. Several dynein genes were expressed at high levels in never smokers in our study, consistent with the predominance of ciliated cells in our samples. The level of expression of various dynein genes, and therefore the balance of cell types being sampled, did not change in smokers. This finding is consistent with a previous study of antioxidant gene expression in airway epithelial cells from never and current smokers that showed no change in histologic types of cells obtained from smokers (8). Our findings that drug metabolism and antioxidant genes are induced by smoking in airway epithelial cells is consistent with *in vitro* and *in vivo* animal studies (summarized in ref. 9). The high-density arrays used in our studies allowed us to define the effect of cigarette smoking on a large number of genes not previously described as being affected by smoking.

Effects of Smoking Cessation. Relatively little information is available about how smoking cessation alters the effects of smoking on airways. Cough and sputum production decreases rapidly in smokers with bronchitis who cease to smoke (23). The accelerated decline in forced expiratory volume, which characterizes smokers with chronic obstructive pulmonary disease, reverts to an age-appropriate decline of forced expiratory volume when smoking is discontinued (24). However, the allelic loss in airway epithelial cells obtained at biopsy changes relatively little in former smokers, and the risk for developing lung cancer remains high for at least 20 years after smoking cessation (6).

Fig. 2A shows a multidimensional scaling plot of never and current smokers according to the expression of the 97 genes that distinguish current smokers from never smokers. Fig. 2B shows that former smokers who discontinued smoking <2 years before this study tend to cluster with current smokers, whereas former smokers who discontinued smoking for >2 years group more closely with never smokers. Hierarchical clustering of all 18 former smokers according to the expression of these same 97 genes also reveals two subgroups of former smokers, with the length of smoking cessation being the only clinical variable that was statistically different between the two subgroups (see Fig. 8, which is published as supporting information on the PNAS web site). Reversible genes were predominantly drug-metabolizing and antioxidant genes.

Thirteen genes did not return to normal levels in former smokers, even those who had discontinued smoking 20–30 years before testing ($P < 9.8 \times 10^{-4}$; threshold determined by permutation analysis). These genes include a number of poten-

tial tumor suppressor genes, e.g., TU3A and CX3CL1, which are permanently decreased, and several putative oncogenes, e.g., CEACAM6 and HN1, which are permanently increased (see Fig. 3 and Table 9, which is published as supporting information on the PNAS web site). Three metallothionein genes remain decreased in former smokers. Metallothioneins have metal-binding, detoxification, and antioxidant properties and have been reported to affect cell proliferation and apoptosis (25). The metallothionein genes that remained abnormal in former smokers are located at 16q13, suggesting that this may represent a fragile site for DNA injury in smokers. The persistence of abnormal expression of select genes after smoking cessation may provide growth advantages to a subset of epithelial cells, allowing for clonal expansion and perpetuation of these cells years after smoking had been discontinued. These permanent changes might explain the persistent risk of lung cancer in former smokers.

Conclusions. We have, for the first time, characterized the genes expressed and, by extrapolation, defined the functions of a specific set of epithelial cells from a complex organ across a broad cross section of healthy individuals. Large-airway epithelial cells appear to serve antioxidant, metabolizing, and host-defense functions. Cigarette smoking, a major cause of lung disease, induces xenobiotic and redox-regulating genes and several oncogenes and decreases expression of several tumor suppressor genes and genes that regulate airway inflammation. We also identified a subset of smokers who respond differently to cigarette smoke and may be predisposed to its carcinogenic effects. Finally, we have explored the reversibility of altered gene expression when smoking was discontinued. The expression level of smoking-induced genes among former smokers began to resemble that of never smokers after 2 years of smoking cessation. Genes that reverted to normal within 2 years of cessation tended to serve metabolizing and antioxidant functions. Several genes, including potential oncogenes and tumor suppressor genes, failed to revert to never-smoker levels years after cessation of smoking. These later findings may explain the continued risk for developing lung cancer many years after individuals have ceased to smoke. In addition, results from this study raise the possibility that the airway gene expression profile in smokers may serve as a biomarker for lung cancer.

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